



PROPOSAL

HACKELITE 2.0

TEAM NAME	B2
UNIVERSITY	University Of Moratuwa
DOMAIN	Social Impact



University of Moratuwa
IEEE Student Branch

PROBLEM IDENTIFICATION AND CONTEXT

The esports industry is growing rapidly and becoming a multi-billion-dollar global market. However, many tournaments—especially at the grassroots and semi-professional levels—still rely on manual and outdated systems. This creates several key problems that affect players, organizers, and fans:

- **Lack of Transparency:** Tournament draws and brackets are often managed by a single person without oversight, raising concerns about fairness and possible bias.
- **High Administrative Effort:** Organizers spend a lot of time creating brackets, updating results, and managing communication, which takes focus away from improving the event itself.
- **Weak Spectator Experience:** Fans often receive slow or limited updates, usually through static images or basic posts, which makes following a tournament less exciting and engaging.
- **Information Gaps:** After a match, there's usually no quick summary or highlight, making it hard for fans to catch up on important moments.

These challenges reduce trust in the tournament process and limit the overall quality and growth of esports events.

Project Aegis aims to solve these issues by creating a modern, transparent, and efficient platform for managing esports tournaments—benefiting organizers, players, and spectators alike.

SOLUTION CONCEPT AND DISTINCTIVENESS



SEAMLESS AUTOMATION

Automating key processes like notifications and match summaries to reduce organizer workload



VERIFIABLE FAIRNESS

Utilizing blockchain technology to generate tamper-proof tournament draws.



ENHANCED ENGAGEMENT

Providing a live, dynamic, and shareable viewing experience for all stakeholders

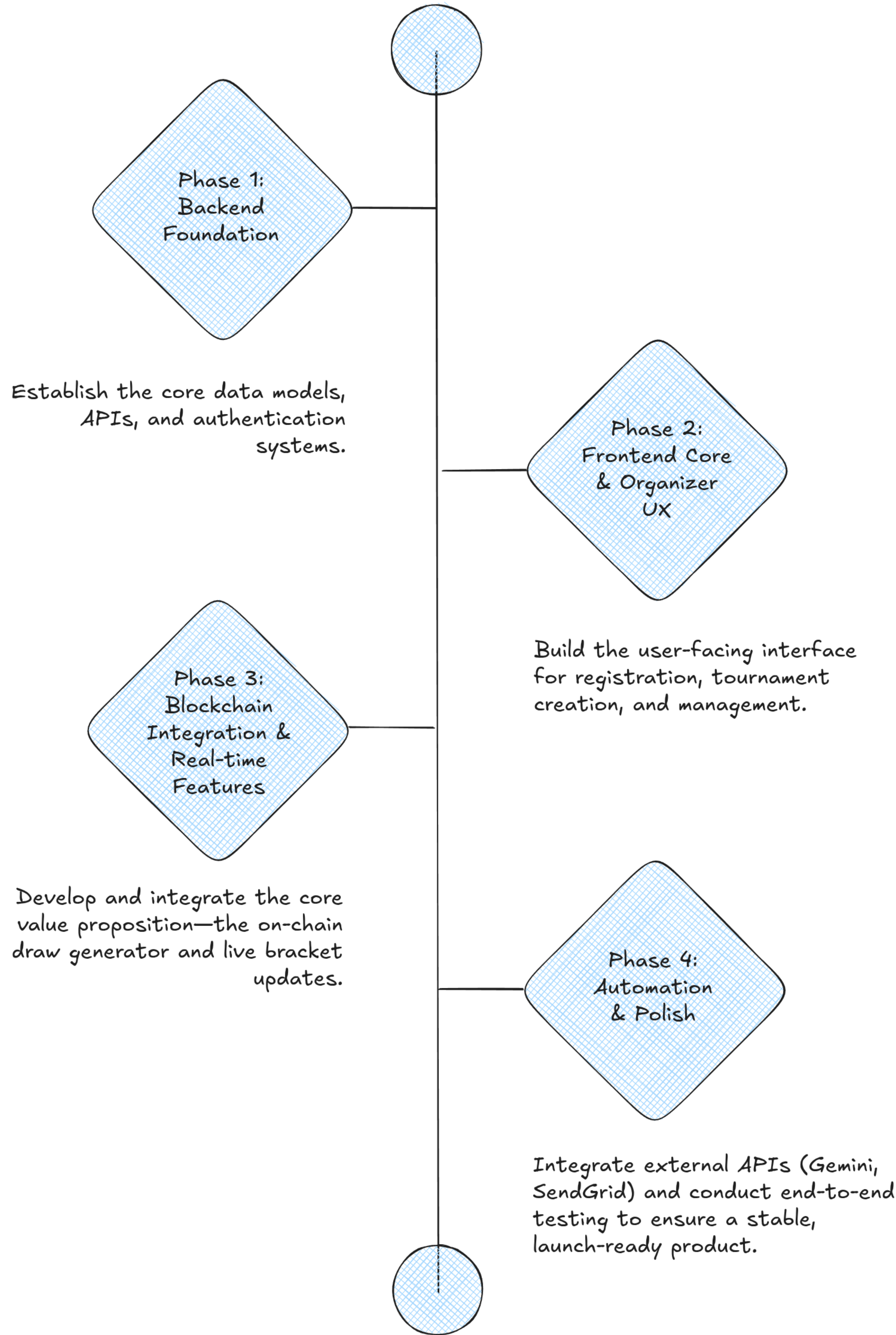
Project Aegis is a web-based platform designed to facilitate the creation and management of esports tournaments with an emphasis on cryptographic-proof and user experience.

The Minimum Viable Product (MVP) will deliver a complete, end-to-end flow for tournament organizers and participants.

SOLUTION CONCEPT AND DISTINCTIVENESS

Development Methodology

We will adopt an Agile, feature-driven methodology to deliver the MVP within a 10-week timeline. The project is broken down into four distinct phases, ensuring iterative progress and the ability to adapt:



SOLUTION CONCEPT AND DISTINCTIVENESS

Project Aegis differentiates itself from existing tournament platforms in two key ways:

- **Provably Fair Draws (Blockchain Integration):** While other platforms claim fairness, Aegis will provide cryptographic proof. By generating the tournament draw via a smart contract on the Polygon blockchain, the process becomes transparent, auditable, and immutable. Any participant can independently verify that the draw was not tampered with. This is our core competitive advantage.
- **AI-Powered Narrative Generation (NLP Integration):** Leveraging the Google Gemini API, Project Aegis will automatically generate exciting, narrative summaries of completed matches based on simple score inputs. This unique feature enhances the user experience for fans and provides valuable, shareable content that no other platform currently offers in an automated fashion.

VALUE PROPOSITION AND MARKET STRATEGY

Value Proposition

Project Aegis offers distinct value to each of its key user segments:

1. For Tournament Organizers:

- Legitimacy & Trust: Attract more participants by offering verifiably fair tournaments.
- Efficiency: Automate draw generation, notifications, and content creation, saving significant time.
- Professionalism: Provide a polished, real-time bracket experience for players and fans.

2. For Players & Teams:

- Guaranteed Fairness: Compete with confidence, knowing the bracket is tamper-proof and transparent.
- Clarity: Receive timely, automated notifications about schedules and results.

3. For Fans & Spectators:

- Real-time Engagement: Follow the action on a live-updating bracket, eliminating the need for page refreshes.
- Rich Content: Enjoy AI-generated summaries that capture the excitement of each match.

VALUE PROPOSITION AND MARKET STRATEGY

Market Strategy

Our initial strategy is focused on validating the MVP and achieving product-market fit within the grassroots and semi-pro esports communities.

- **Internal Validation:** Execute the "Launch Readiness" objective by running at least 5 pilot tournaments internally to ensure end-to-end stability.
- **Targeted Beta Launch:** Partner with a select group of community tournament organizers (e.g., for games like Valorant, League of Legends, Counter-Strike) to use the platform for free. This will provide invaluable feedback and initial case studies.
- **Community-Led Growth:** Leverage the platform's unique "shareable bracket URL" feature. As organizers share their tournament links on social media and community forums (like Discord and Reddit), it will organically expose the platform to players and fans.
- **Feedback-Driven Iteration:** Collect user feedback from the beta phase to prioritize the "Post-MVP" feature backlog (e.g., AI-powered skill ratings, advanced tournament formats), ensuring that V1.1 directly addresses user needs.

The Final Product will be able to support to have Premium Competitions with a closed invitation system, which will need to have a Premium Account for the Competition Organiser and the players.

TECHNICAL FRAMEWORK AND DEVELOPMENT PLAN

Project Aegis is built using a modern, scalable, and modular tech stack to ensure performance, reliability, and rapid development.

Frontend:

Built with Next.js, enabling fast and dynamic user interfaces.

Backend:

Powered by FastAPI (Python) for high-speed, asynchronous API handling.

Database:

Uses PostgreSQL, a robust relational database for structured data.

Blockchain Integration:

Runs on Polygon for low-cost, fast transactions.

Smart contracts are developed in Solidity, using Hardhat for deployment.

Ethers.js handles blockchain interaction on the frontend.

External APIs:

Google Gemini API for AI-generated match summaries.

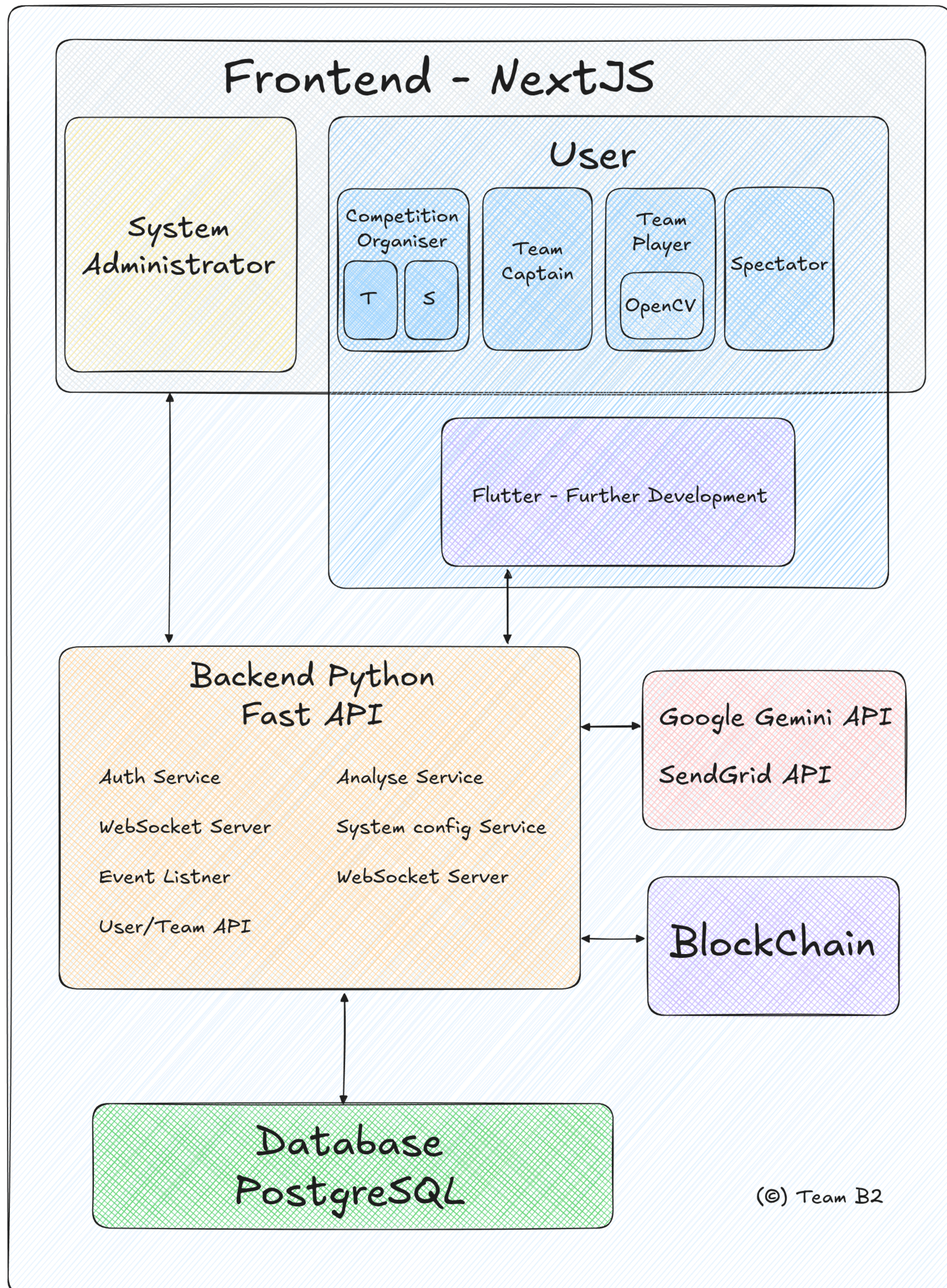
SendGrid/Mailgun for automated email notifications.

Deployment:

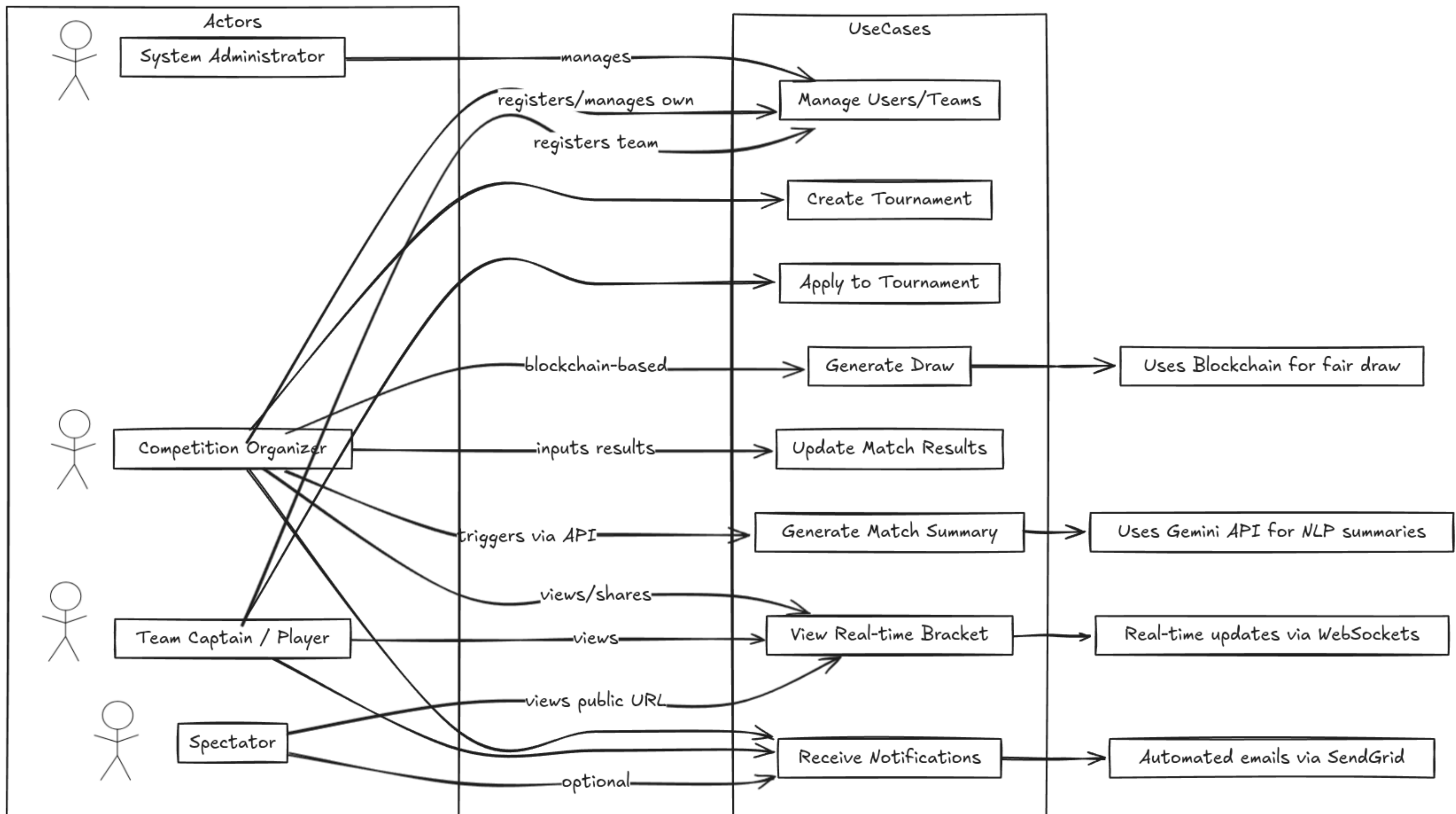
Frontend on Netlify, backend and database on Azure.

This stack ensures a secure, transparent, and seamless tournament experience across all user segments.

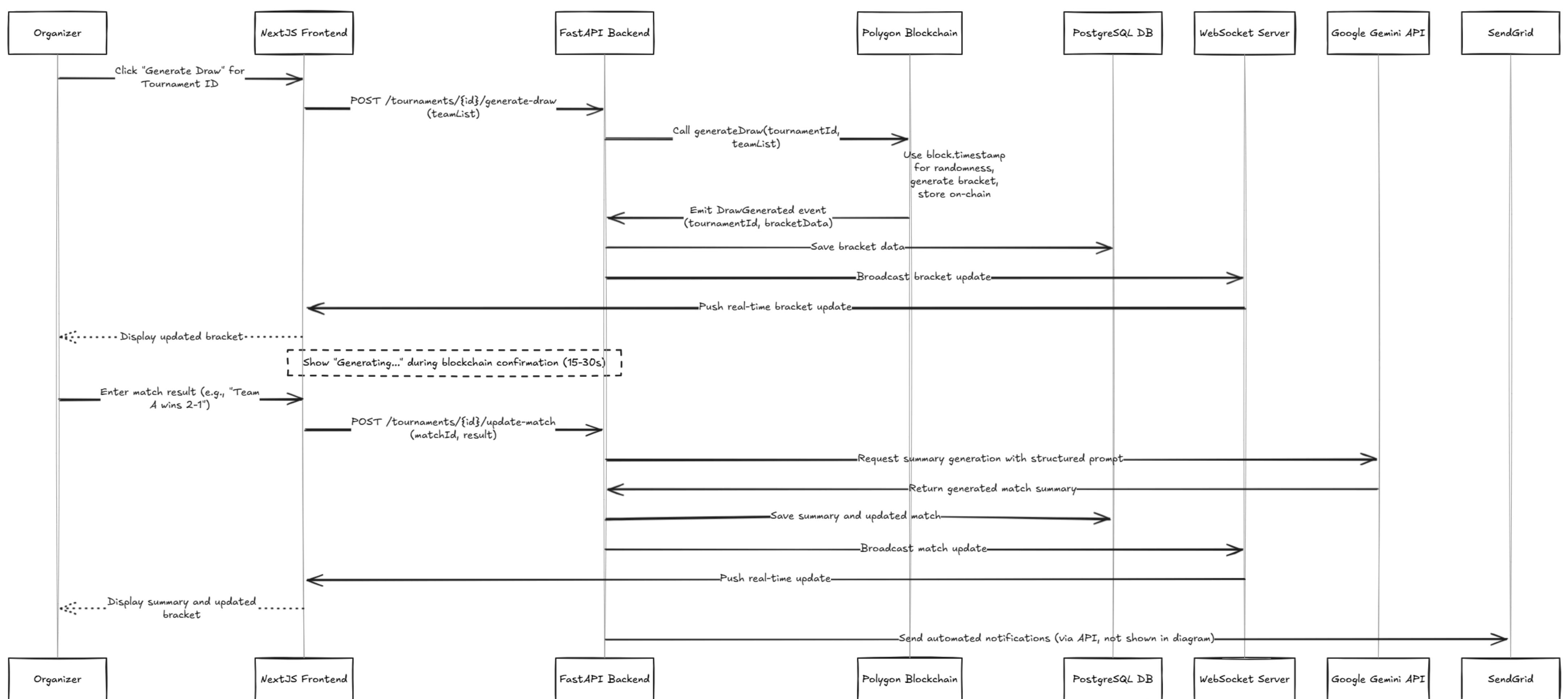
ARCHITECTURE DIAGRAMS/TECHNICAL DIAGRAMS



USER INTERACTION AND APPLICATION SCENARIOS



Use Case Diagram for Project Aegis MVP



FIGMA WIREFRAMES & MOCKUPS

[Click here to view](#)

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